

HATCHSIDE · PROBLEM STATEMENT

HatchSide problem statement

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; safety section added; first candidate partners recorded
0.2	2026-10-03	Amish Chadha	Populated to TRL 2; budget worded as a value-engineering target; open questions answered (HTS-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

HatchSide problem statement

Cleaning, sampling and retrieval jobs in sewers, tanks and digesters are still done by sending a person inside. The air in these spaces can kill in seconds, and it kills the rescuers too.

■ The problem

Deaths follow a pattern: one worker enters to clean or free something, collapses, and others follow. In a 1989 Michigan case, five family members died one after another in a manure pit ([NIOSH FACE 89-46](#)). In India, most sewer and septic tank victims audited had no safety equipment ([The Wire, 2025](#)), and in the US repeated tank cleaning deaths followed a failure to test the atmosphere before entry ([OSHA, 2024](#)).

Existing tools cover parts of the job. The Bandicoot robot cleans manholes with a pneumatic arm, but beta units cost about INR 10 to 12 lakh to make ([Forbes India](#)). Commercial confined-space tripods with winches cost about USD 5,850 to 6,120 and are built to lower and retrieve people ([PK Safety](#)). Sludge samplers exist but work alone ([Hach](#)). What is missing is a low-cost, open frame that carries many non-entry tools, so that the default for routine jobs becomes staying outside.

■ Users and context

User	Need	Context
Sanitation and septic tank workers	Clear, sample and retrieve without going in	Street manholes and household tanks, often informal work with no gear
Municipal and utility crews	A portable, fast-setup kit for routine desilting and inspection	Sewer networks, pump stations and drains
Farm families and biogas technicians	Service and reseal a household digester from the opening	Rural fixed-dome digesters
Oil, gas and industrial tank contractors	Sample and loosen residues before any entry decision	Frac tanks, storage tanks and tankers

■ Operating environment

- Street manholes about 0.6 m (24 in) across and tank hatches on roofs or sides; depths to about 10 m (33 ft) assumed for a first version (estimate).
- Toxic, flammable or oxygen-deficient atmospheres inside the space, including hydrogen sulfide, methane and carbon monoxide.
- Traffic on streets, uneven ground, mud and sewage splash at the rim.
- Outdoor use in heat and monsoon rain; corrosive sewage and sludge on every wetted part.
- Crews of two to four people, with a handcart or small vehicle for transport.

■ Constraints

- Value-engineering target of USD 5,000 for the tripod, winch and the five tool heads including DomeReach (a hypothetical control target, not a spending limit).

- Hand winch only; no engine at the hatch.
- Kit carried in loads of 25 kg (55 lb) or less each (target).
- Every tool head worked from outside the space; no design step may require entry.
- Open design: hardware under CERN-OHL-S-2.0, any software under MIT.

■ Out of scope

- Planned human entry and the equipment that supports it (breathing apparatus, entry permits).
- Powered robots and articulated robotic arms.
- Gas detection electronics, which crews should buy as certified instruments.
- Sewer jetting and vacuum trucks.
- Building or repairing digester structures beyond sealing from the opening.

■ Prior work

Prior work	What it does	Gap for these users	Source
Genrobotics Bandicoot	Pneumatic, remote-controlled manhole cleaning robot with a sweeping arm	High cost and a powered robot; the preliminary screen notes three patents and ongoing litigation around its arm and gripper	link
3M DBI-SALA confined space tripod and winch	Aluminium tripod of 2.13 or 2.74 m with a retrieval winch for entry and rescue	Built for lowering people in; costs about USD 5,850 or more and carries no work tools	link
US5431248A confined space lowering and retrieving apparatus	Tripod or quadruped frame for lowering and retrieving personnel through manways	Expired patent for a personnel system; free prior art for a frame	link
Sludge Judge sampler	Core sampler for settled solids depth in tanks	Single task and hand-held; limited depth and no frame	link
Biogas plant maintenance guidance	Guidance that digester maintenance should be done from above ground where possible	Guidance only; no tools for small household digesters	link

■ Co-design

A sanitation workers' organisation or municipal sanitation department that already runs mechanised cleaning programmes, plus a biogas programme with field technicians for DomeReach, so heads are shaped by the real jobs and real openings.

First candidates to approach (none agreed): a municipal sewer department in India running mechanised cleaning under the national mechanised sanitation programme, and a household biogas programme with field technicians in India or Nepal (HTS-DDR-001, D11).

Co-design checklist:

- [] Confirm the five tasks with crews who do them today.

- [] Measure common manhole, tank hatch and digester openings in the first region.
- [] Walk through each head at a mock opening with the crew.
- [] Confirm that crews will carry a certified gas detector.

■ Safety

Safety: HatchSide lifts loads over an open confined space whose air can kill in seconds. It is an open engineering reference for a material-handling frame, never certified rescue, lifting or fall-protection equipment. Nobody enters the space, and the winch never lifts or lowers a person. Test the air at the opening with a certified gas detector before and during work, keep sparks and flames away from sewers and digesters, guard the opening from traffic and falls, and never stand under a raised load.

■ Answered questions

The scaffold's open questions were answered at the TRL 2 review (HTS-DDR-001) and are tracked in the design decisions register (HTS-DEC-001):

- Tasks that still force entry: none of the five routine tasks needs entry with the five heads; any task found in co-design that does gets a new head, not an entry.
- The retrieval head stays, as a clip-on guide only; the haul is made with certified rescue equipment.
- The frame carries 150 kg; WellSling's 400 to 600 kg loads need their own frame.
- DomeReach uses a generic roller-applied digester sealant, named by type.
- First trial hosts are recorded as candidates to approach, above.